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Roll No

CE-7005(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Earthquake Resistant Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Discuss Indian code provisions for seismic force calculation in a 3 storey RCC framed in fill building on soft soil.
2.
 - a) Explain briefly measurement of earthquakes.
 - b) Distinguish between
 - i) Body waves and surface waves
 - ii) Lithosphere and asthenosphere
3.
 - a) Explain the use of response spectrum in earthquake resistant design.
 - b) Derive an expression for systems with single Degree of freedom.
4.
 - a) Discuss ductility requirement in RCC building for earthquake resistance.
 - b) Discuss load resisting structural systems.

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5. Explain the base isolation and describe the design principle of base isolation of structures.
6.
 - a) Explain the analysis of infill wall using equivalent diagonal strut for infill without opening.
 - b) Discuss the various objectives of Seismic Design.
7.
 - a) Derive general equation of motion for undamped free vibration of an single degree of Freedom System.
 - b) Explain seismograms and accelerograms.
8. Define the following:
 - a) Retrofitting of structures
 - b) Seismic control systems
